We'll use ACME's journey to illustrate key ideas. As we examine Cedar's structure, its underlying PARC model (Principals, Actions, Resources, Conditions), and its evaluation logic, you'll see how policies are created, scoped, and put into action. By the end of this chapter, you'll have a clear understanding of how dynamic access control is represented in policies and a foundation for the more advanced topics to come.

7.1 Introducing Cedar

Cedar is a purpose-built policy language designed for fine-grained access control. Amazon Web Services (AWS) developed it to power services like Amazon Verified Access and Amazon Verified Permissions (AVP), and it is now an open-source project. Unlike general-purpose languages adapted for authorization, Cedar was designed specifically to clearly and safely express real-world access control logic.

At ACME Corp, the identity team chose Cedar for the same reasons many engineering teams do: It offers the right balance of power, performance, and auditability. As a policy language, Cedar has four main strengths:

- **Expressive**: Cedar can model complex authorization logic using attributes, relationships, and environmental conditions. It works well across various policy styles, whether you're assigning access based on job roles, relationships within an organizational graph, or dynamic attributes such as device trust or time of day.
- **Performant**: Since access decisions often occur in the critical path of a request, speed is vital. Cedar is optimized for real-time evaluation, providing predictable performance that makes it suitable for large-scale production environments.
- **Analyzable**: Authorization systems must be secure, not just fast. Cedar policies are designed to be statically analyzable, meaning you can reason about them before they are executed. You can verify if a policy is satisfiable, if it might grant unintended access, or if a modification causes regressions—all without deploying it first.
- **Open**: Cedar is open source and designed for embedding. You can run it as a library within your services, integrate it into custom policy